

## Do it right, get recognised

A mandatory all-green construction system will take time, and this being a transition phase, it would be not advisable to use the regulatory mandatory card, as it would invite more criticism than participation. The green movement needs to grow, and grow thick and fast to a point where making it mandatory wouldn't pinch a soul. We need to take concrete steps in this direction, but cautiously.

Although not mandatory, there are a few certifications and guidelines available in India: 1. Builders can voluntarily apply for The Leadership in Energy and Environmental Design (LEED) rating available in India. There are around 500 registered LEED buildings in India, and around 50 have already been certified. The LEED guidelines are also being tweaked to suit the Indian conditions.

2. Other voluntary rating systems such as those by The Energy and Resources Institute (TERI) and Eco-Housing India are also available, although the TERI rating is still in the development stages.

3. The Ministry of New and Renewable Energy, in collaboration with TERI, has set up the Green Rating for Integrated Habitat Assessment (Griha)—a rating agency which will approve a building as green under certain parameters.

4. BEE star labels for factories and offices are becoming a norm where energy consumption is above 500 kVA.

There are several steps that the builder and occupants can take to ensure a green building. I would suggest rainwater harvesting, a recycled water plant, and a solar and/or wind hybrid mechanism—in principle, a major focus on water and power. There are many more elements like materials, environmental impact and efficiency of various systems that have a deep impact, but water and power are the ones where the impact and savings are instantaneously visible.

—Nilesh Gandhi, architect and specialist in sustainable buildings, on certifications and regulations

Of late, a few solar-powered gadgets—watches, mobile battery chargers, etc.—are also starting to appear in the market. These are definitely worth a try.

Household appliances are becoming even smarter. Though not commercially available yet, GE has developed appliances that can respond to signals from an electric utility that demand is high. For example, a dryer might shift from using more than 4 kilowatts of energy to just 700 watts, drying clothes at a slower place and decreasing demand from the grid. Plus, a 'home energy manager' device will allow consumers to set multiple appliances to automatically respond to signals from the utility. The ability to reduce peak demand could allow utilities to put off building new power plants. According to a GE spokesperson, if 250,000 dryers responded to a signal from the utility, it would be enough to offset all the power generated by a coal power plant!

Companies need to do more than homes: go for cooler, energy-efficient servers, sustainable design of data

centres, motion-sensors to switch off HVAC and lighting systems when there is nobody in the room, and so on.

Ananthaswamy suggests that offices can even go in for green-mode power supply controllers for computers. "These make sure that the computer monitors go into sleep mode when not in use and help consume less energy. The decision-making is shared and if one forgets to turn off the monitors when they are not in use, green-mode power controllers, along with system controllers, turn the monitors off," he explains.

Another area where one can save energy is by using efficient uninterruptible power supply (UPS) units. We have a lot of indigenous innovations in this space. For example, there is a high-capacity UPS system for enterprises, designed by Uniline, that uses a high-frequency pulse-width-modulated rectifier/inverter system based on insulated-gate bipolar transistors. By doing this, they have managed to achieve an input power factor (PF) greater than 0.99 and a total harmonic distortion less than five per cent. High input power factor helps

to reduce the power consumption by as much as 30 per cent. Also, the systems are designed for hot Indian climate and do not even require air-conditioning—another power saver!

## Let the Sun in!

Every commercial building's rooftop has a large unused area. Solar panels can be installed and used for a variety of purposes ranging from water heating to lighting. They can be used especially to power areas of a building that are lit by LEDs. "I say this because the solar panels' light-to-electricity conversion efficiency is low and even a large panel will not produce large amounts of power. However, the power produced by these panels is more than adequate to power LED lamps. An inverter with over 95 per cent efficiency can be used to convert the solar power into usable alternating current (AC)," says Ananthaswamy.

Ananthaswamy goes on to explain that one can benefit from a solar panel for thirty years. The barrier to entry today is the cost of the panels, sensors, monitors, etc. However, if you amortise the cost over thirty years, it ends up being inexpensive. In the last six to eight months, the cost-per-watt of solar power has come down 30-40 per cent. With the National Solar Mission and other government subsidies, solar power will become affordable in the near future. The breakeven for solar panels until recently used to be seven years, but now it is just around three years. Added solar panel capacity and the wider adoption of renewable energy through awareness will bring the breakeven point even lower.

What is more, solar energy does not have to be tapped only from the rooftops. According to recent news reports, Dow Jones, the media giant, is going in for an unprecedented solar panel installation. The 4.1-megawatt solar power system will be installed at its 200-acre campus in South Brunswick, New Jersey, next year, and cover nearly 21,368 sq. m of parking space. This is an excellent example of solar